



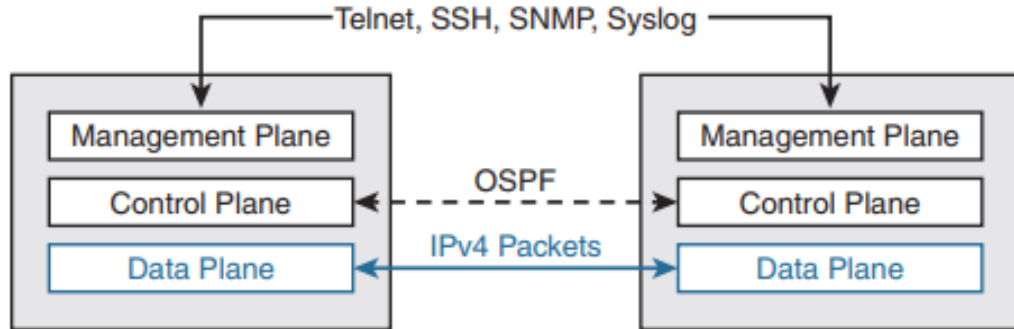
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Protocols necessary to maintain the network

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Planes in networking

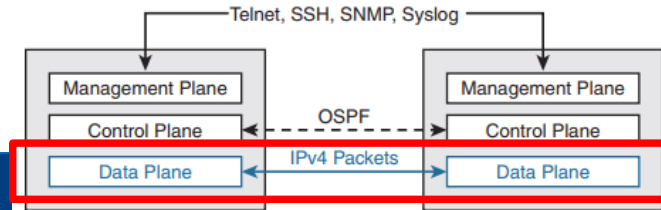
Planes in networking



Data plane

Data plane:

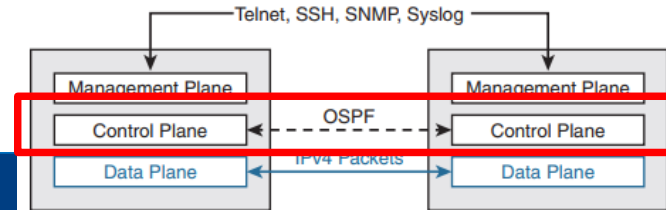
- Transferring data as quickly as possible,
- Protocols:
 - Layer 1, 2 Encapsulation and Decapsulation (e.g. creating an 802.1Q trunk header)
 - IP routing table and MAC address table lookup
 - Message filtering (e.g. ACL lists or port security)
 - Data encryption by adding a new IP header (e.g. VPN)
 - IP address swapping (e.g. NAT)



Control pane

Control pane:

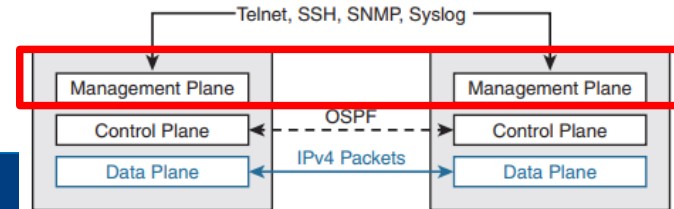
- Enables and maintains data transfer over the network
- Protocols:
 - Routing (OSPF, EIGRP, RIP, BGP),
 - ARP in IPv4,
 - STP.



Management plane

Management plane:

- The management plane is responsible for all functions that are not directly related to controlling the data plane
- Protocols:
 - Telnet, SSH, SNMP, Syslog

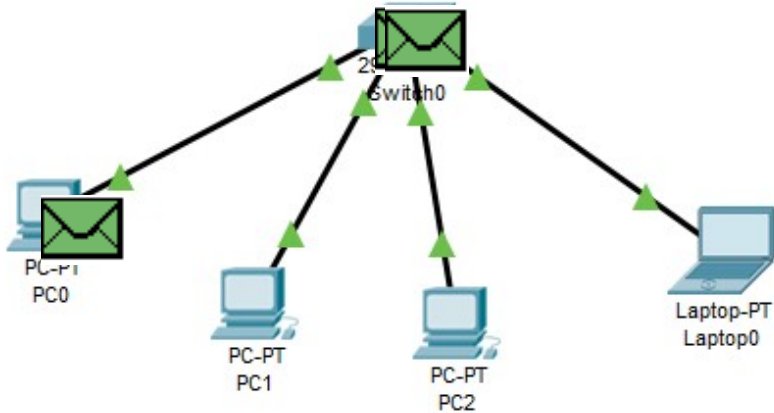


ARP in IPv4

Network protocols

		MODEL OSI	PDU	Addres	Device	Protocols	MODEL TCP
Implementation	Software	7 Application				HTTP, DNS,	
		6 Presentation				SMTP, POP,	Application
		5 Session				SSL, SQL, DHCP	
		4 Transport	Segment TCP Datagram UDP	Port Number		TCP, UDP, SCTP	Transport
	hardware	3 Network	Packet	IP address	Router (Ruter)	IPv4, IPv6, ICMP	Internet
		2 Link layer	Frame	Physical address MAC	Switch (Przełącznik)	ARP, Ethernet, PPP, CDP	Network
		1 Physical			Hub (Koncentrator)	Bluetooth, DSL, 802.11	Access

Steps to get the MAC address



ARP

The ARP protocol provides two basic functions:

- Mapping IPv4 addresses to MAC addresses
- Maintaining a mapping in ARP table

ARP table

- an array in RAM that connects network layer (IPv4) addresses with host hardware (MAC) addresses
- is stored in layer 3 devices (network)

Description of ARP messages

8 bit		8 bit	8 bit	8 bit
HARDWARE TYPE: 0x0001		PROTOCOL TYPE: 0x0800		
HLEN: 0x06 Dł. adresu sprzętowego (MAC)	PLEN: 0x04 Długość adresu protokołu (IP)	OPCODE: 0x0002 operacja		
SOURCE MAC: 000A.4136.B2B5		SOURCE IP :192.168.0.1		
		TARGET MAC:000A.4114.AA8A		
TARGET IP:192.168.0.2				

ARP proxy (ARP pośredniczące)

APR proxy - a service on the router where the router sees an ARP request that cannot reach a given host, and the router knows the route leading to it, then the router replies on behalf of this host, providing its own MAC address in the ARP response.

ARP poisoning, ARP spoofing (Zatruwanie ARP)

ARP poisoning - an attack technique on an Ethernet network; consists in sending false ARP messages to the Ethernet LAN. These frames contain bogus MAC addresses that mislead network devices such as switches. As a result, frames destined for one node may be sent to another by mistake.

RARP

RARP (ang. Reverse Address Resolution Protocol) – a communication protocol that allows you to convert 48-bit physical MAC addresses to 32-bit IP addresses

Display the ARP table of the router and the computer

- Router> show arp

```
Router>
Router>
Router>show arp
Protocol  Address          Age (min)  Hardware Addr  Type   Interface
Internet  192.168.20.2      -          00D0.FF6B.CE01  ARPA   FastEthernet0/0
Router>
Router>
```

```
ubuntu@prod5:~$
ubuntu@prod5:~$
ubuntu@prod5:~$ arp -a
? (10.144.204.93) at 96:7d:5f:f5:f5:8d [ether] on ztbt06b7xf
? (10.144.3.167) at <incomplete> on ztbt06b7xf
? (10.144.41.245) at 96:d5:2f:ca:86:f7 [ether] on ztbt06b7xf
? (192.168.136.21) at <incomplete> on wlan0
_gateway (192.168.136.1) at 00:10:db:ff:10:00 [ether] on wlan0
ubuntu@prod5:~$
ubuntu@prod5:~$
ubuntu@prod5:~$
```

```
Wiersz polecenia
C:\Users\user>arp /a

Interface: 192.168.1.101 --- 0x5
Internet Address      Physical Address      Type
192.168.1.1           94-0c-6d-aa-d8-50     dynamic
192.168.1.255         ff-ff-ff-ff-ff-ff     static
224.0.0.2             01-00-5e-00-00-02     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.252           01-00-5e-00-00-fc     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.246.1 --- 0xb
Internet Address      Physical Address      Type
192.168.246.254       00-50-56-fa-80-4e     dynamic
192.168.246.255       ff-ff-ff-ff-ff-ff     static
224.0.0.2             01-00-5e-00-00-02     static
224.0.0.22            01-00-5e-00-00-16     static
```

ARP w linux

Obsolete comannds

arp -help

arp -a

wyświetlanie

arp -n

wyświetlanie

sudo arp -d __.__.__.__

kasowanie adresu

Recomennded commands:

sudo ip neigh show

sudo ip -s neigh flush all # kasowanie wszystkich

ICMPv4

Network protocols

		MODEL OSI	PDU	Addres	Device	Protocols	MODEL TCP
Implementation	Software	7 Application				HTTP, DNS,	Application
		6 Presentation				SMTP, POP,	
		5 Session				SSL, SQL, DHCP	
		4 Transport	Segment TCP Datagram UDP	Port Number		TCP, UDP, SCTP	Transport
	hardware	3 Network	Packet	IP address	Router (Ruter)	IPv4, IPv6, ICMP	Internet
		2 Link layer	Frame	Physical address MAC	Switch (Przełącznik)	ARP, Ethernet, PPP, CDP	Network Access
		1 Physical			Hub (Koncentrator)	Bluetooth, DSL, 802.11	

ICMP

Bit 0 - 7	Bit 8 - 15	Bit 16 - 23	Bit 24 - 31
Typ	Kod	Suma kontrolna	

Type	Description ICMP Message Types
0	Echo Reply (Ping Reply, used with Type 8, Ping Request)
3	Destination Unreachable
4	Source Quench
5	Redirect
8	Echo Request (Ping Request, used with Type 0, Ping Reply)
9	Router Advertisement (Used with Type 9)
10	Router Solicitation (Used with Type 10)
11	Time Exceeded
12	Parameter Problem
13	Timestamp Request (Used with Type 14)
14	Timestamp Reply (Used with Type 13)
15	Information Request (obsolete) (Used with Type 16)
16	Information Reply (obsolete) (Used with Type 15)
17	Address Mask Request (Used with Type 17)
18	Address Mask Reply (Used with Type 18)

ICMP - TYPE 3 (DESTINATION UNREACHABLE)

```
C:\Users\pawel.krol>ping 192.168.179.240

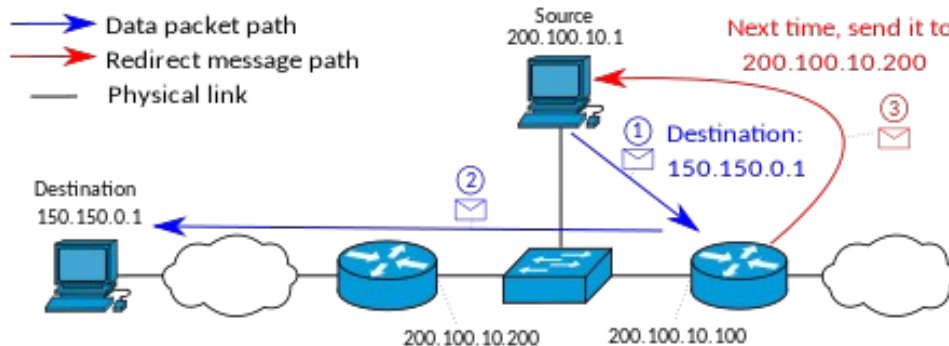
Pinging 192.168.179.240 with 32 bytes of data:
Reply from 192.168.179.217: Destination host unreachable.
Reply from 192.168.179.217: Destination host unreachable.
Reply from 192.168.179.217: Destination host unreachable.
Reply from 192.168.179.217: Destination host unreachable.

Ping statistics for 192.168.179.240:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

C:\Users\pawel.krol>
```

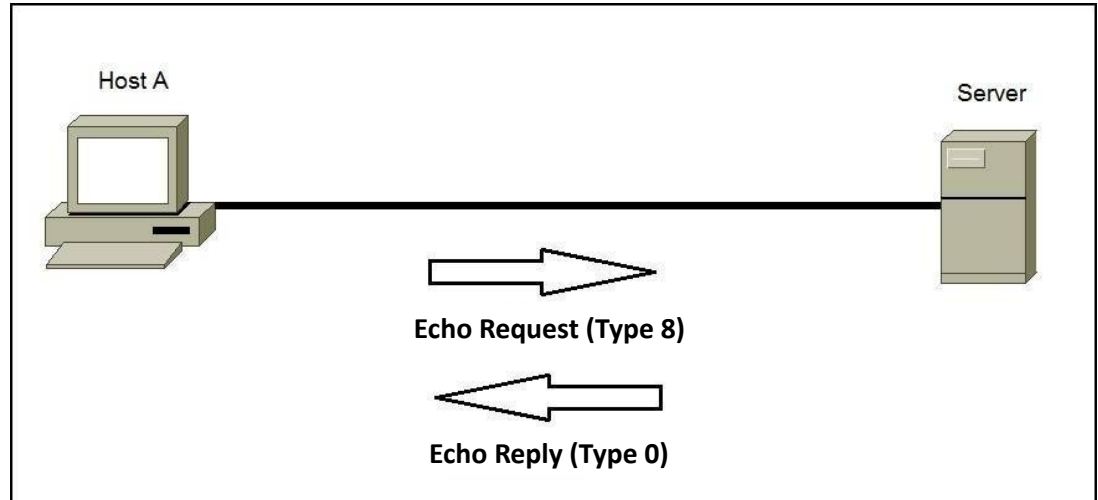
ICMP - TYPE 5 (Redirect Message)

- Redirect requests data packets be sent on an alternative route.
- The message informs a host to update its routing information (to send packets on an alternative route).



ICMP - TYPE 8, 0 (REQUEST – REPLY)

- Implementowane w „ping”



DHCPv4

Network protocols

		MODEL OSI	PDU	Addres	Device	Protocols	MODEL TCP
Implementation	Software	7 Application				HTTP, DNS,	Application
		6 Presentation				SMTP, POP,	
		5 Session				SSL, SQL, DHCP	
	hardware	4 Transport	Segment TCP Datagram UDP	Port Number		TCP, UDP, SCTP	Transport
		3 Network	Packet	IP address	Router (Ruter)	IPv4, IPv6, ICMP	Internet
		2 Link layer	Frame	Physical address MAC	Switch (Przełącznik)	ARP, Ethernet, PPP, CDP	Network Access
		1 Physical			Hub (Koncentrator)	Bluetooth, DSL, 802.11	

BOOTP

BOOTP (ang. Bootstrap Protocol) – a decommissioned communication protocol that allows computers on a network to obtain an IP address from a configuration data server

- The assigned addresses are for 30 days,
- Uses the UDP protocol in the transport layer,
- Only the IP address,
- Susceptible to errors,

DHCP is the expansion and successor to BOOTP.

DHCP Domain Host Name Protocol

DHCP Domain Host Name Protocol - dynamic host configuration protocol.

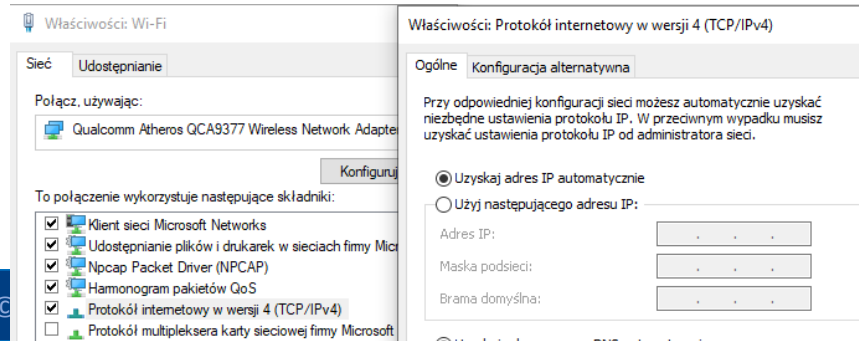
It is implemented in hosts, i.e. in devices that support the ISO / OSI model network layer:

- Computers
- Routers

DHCP Domain Host Name Protocol

A communication protocol that allows hosts to obtain configuration data from the server:

- ❖ host IP address,
- ❖ gateway IP address,
- ❖ DNS server address,
- ❖ subnet masks.



Introduction to DHCP

CLIENT

DHCP-DISCOVER
„I need an IP address”

broadcast

SERWER
DHCP

DHCP-OFFER

„I offer available addresses in the DHCP pool”

DHCP-REQUEST

„I request an address DHCP”

DHCP-ACK

„I confirm the address allocation from DHCP”

DHCP w CISCO – konfiguracja serwera

```
Rxy(config)# service dhcp
```

```
Rxy(config)# ip dhcp excluded-address adres_od [adres_do] !Wykluczenie adresów
```

```
Rxy(config)# ip dhcp pool pula1
```

```
Rxy(dhcp-config)# network 192.168.0.0 255.255.255.0
```

```
Rxy(dhcp-config)# default-router 192.168.0.1
```

```
Rxy(dhcp-config)# domain-name PK
```

```
Rxy(dhcp-config)# dns-server 192.168.0.1
```

```
Rxy(dhcp-config)# lease 5 12 45
```

! lease days hours minutes

```
Rxy(dhcp-config)# end
```

192.168.0.0
192.168.0.95
6, 8, 8, 1111100
 3

DHCP w CISCO – checking reservations

Rxy# show ip dhcp binding

Rxy# show ip dhcp pool

RXY# clear ip dhcp binding *

```
Router#show ip dhcp pool
```

```
Pool pula_dhcp_vlan10 :
```

```
Utilization mark (high/low)      : 100 / 0  
Subnet size (first/next)          : 0 / 0  
Total addresses                   : 62  
Leased addresses                  : 1  
Excluded addresses                : 2  
Pending event                     : none
```

```
1 subnet is currently in the pool
```

Current index	IP address range	Leased/Excl
172.16.0.1	172.16.0.1 - 172.16.0.62	1 / 2

DHCP – client configuration in CISCO router

```
RXY# conf t
```

```
RXY(config)# ip route 0.0.0.0 0.0.0.0 dhcp # default route if needed
```

```
RXY(config)# int fa0/1
```

```
RXY(config-int)# ip address dhcp
```

DHCP – client configuration in CISCO SVI

```
SXY# conf t
```

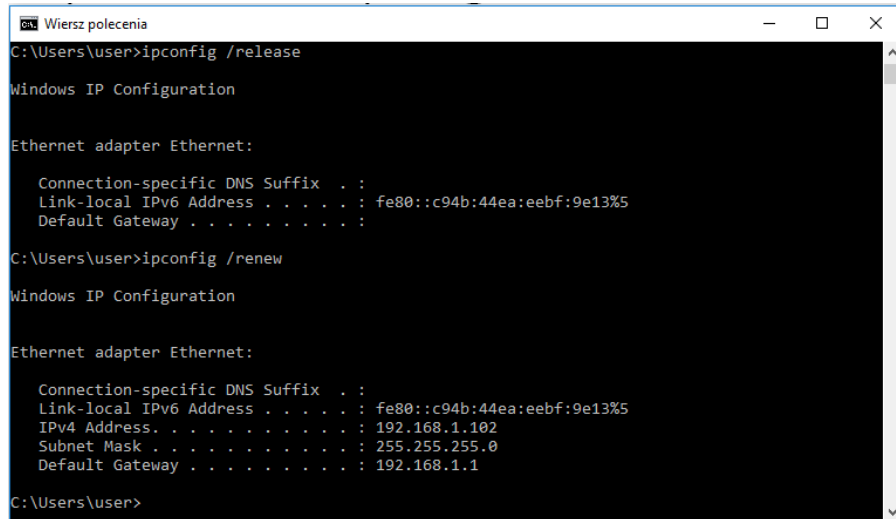
```
SXY(config)# ip route 0.0.0.0 0.0.0.0 dhcp # default route if needed
```

```
SXY(config)# int vlan 1
```

```
SXY(config-int)# ip address dhcp
```

DHCP in Windows

```
C:\> ipconfig
C:\> ipconfig /release
C:\> ipconfig
C:\> ipconfig /renew
```



The screenshot shows a Windows Command Prompt window titled "Wiersz poleceń". The user has entered the following commands: `C:\Users\user>ipconfig /release`, `C:\Users\user>ipconfig /renew`, and `C:\Users\user>`. The output for the first command shows the release of the IPv6 address. The output for the second command shows the renewal of the IPv4 address.

```
C:\Users\user>ipconfig /release

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::c94b:44ea:eebf:9e13%5
    Default Gateway . . . . . : 

C:\Users\user>ipconfig /renew

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::c94b:44ea:eebf:9e13%5
    IPv4 Address. . . . . : 192.168.1.102
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

C:\Users\user>
```

ACL I QOS

ACL access control list

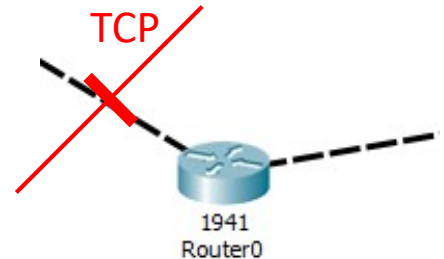
ACL (access control list) contains entries that relate to specific traffic.

The purpose of the ACL is to filter network traffic

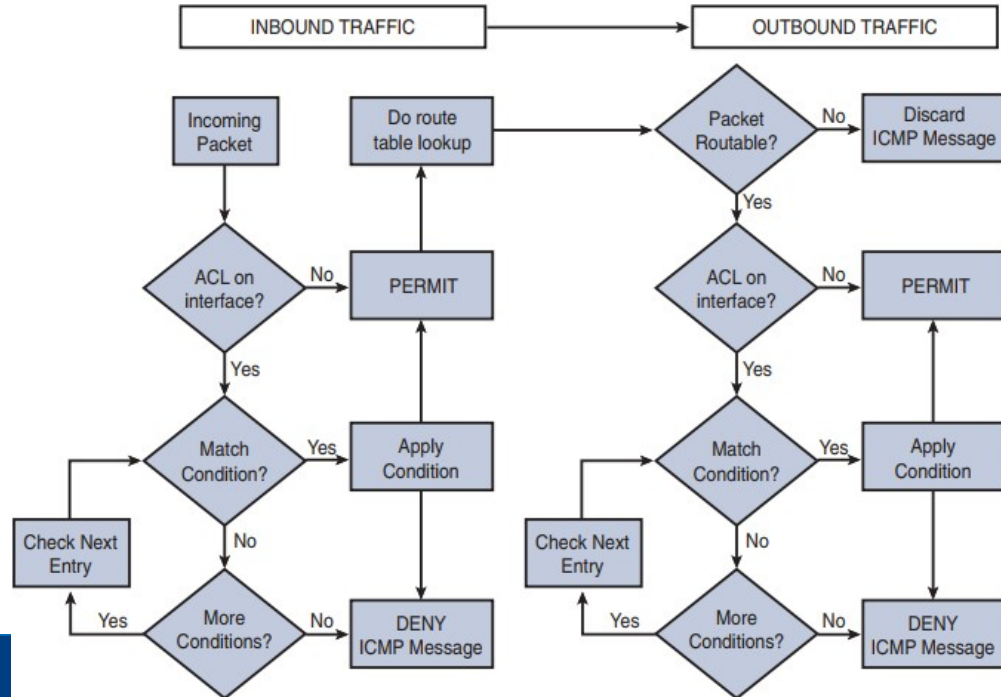
- Permit (passing)
- Deny (blocking)

Filtering can be based on:

- Source and destination IP addresses,
- Protocols,
- Port numbers,



Logic of ACLs



Configuring ACLs in Cisco

ACL configuration steps in CISCO:

1. ACL creation – conditions configuration,
2. Assign an ACL to an interface - either on input or on output

1. ACL creation ACL – types of ACL lists

Standard Numbered	Standard Named	Standard: Matching - Source IP
Extended Numbered	Extended Named	Extended: Matching - Source & Dest. IP - Source & Dest. Port - Others
Numbered: - ID with Number - Global Commands		Named: - ID with Name - Subcommands

1. ACL creation - Cisco numbered lists

Standard (standard ACL):

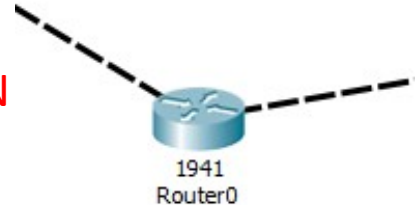
- 1 - 99 **SOURCE**
- 1300 – 1999 **IP ADDRESS**

Extended (feedback ACL):

- 100 – 199 **PROTOCOL** **PORT NUMBER**
- 2000 – 2699 **SOURCE** **DESTINATION**
IP ADDRESS **IP ADDRESS**

Reflective (feedback ACL)

PACKAGES WITH ANSWER ONLY



1. ACL creation – conditions configuration

Numbered list – standard

```
RXY(config)# access-list 1 deny 192.168.1.10
RXY(config)# access-list 1 deny 192.168.1.11
RXY(config)# access-list 1 deny 192.168.1.12
RXY(config)# access-list 1 remark COMMENT TO ACL...
RXY(config)# access-list 1 permit 192.168.1.0 0.0.0.255
RXY(config)# access-list 1 deny any
```

```
Router#sh access-lists
Standard IP access list 1
 10 deny host 192.168.1.10
 20 deny host 192.168.1.11
 30 deny host 192.168.1.12
 40 permit 192.168.1.0 0.0.0.255
 50 deny any
```

1. ACL creation – conditions configuration

Numbered list - extended

```
RXY(config)# access-list 100 deny ICMP host 192.168.2.1 172.16.2.0 0.0.0.255  
RXY(config)# access-list 100 remark COMMENT TO ACL...  
RXY(config)# access-list 100 permit ip any any
```

```
access-list 100 deny/permit  
{gre | icmp | tcp | udp | ip | proto-num}  
{source-ip [wildcard] | host source-ip | any}  
{dest-ip [wildcard] | host dest-ip | any}
```

1. ACL creation – summary

To display ACL lists save in configuration file:

```
RXY(config)# do show running-config
```

```
RXY(config)# do show access-list
```

```
Router#sh acc
Router#sh access-lists
Extended IP access list 100
 10 deny icmp host 172.16.20.2 any (22 match(es))
 20 permit icmp any any (22 match(es))
```

```
!
access-list 1 deny host 192.168.1.10
access-list 1 deny host 192.168.1.11
access-list 1 deny host 192.168.1.12
access-list 1 remark KOMENTARZ DO ACL
access-list 1 permit 192.168.1.0 0.0.0.255
access-list 1 deny any
access-list 100 deny icmp host 192.168.2.1 172.16.2.0 0.0.0.255
access-list 100 remark KOMENTARZ DO ACL
access-list 100 permit ip any any
ip access-list standard NAZWA_ACL
deny host 172.16.2.10
permit any
!
```

2. Assigning ACLs to interfaces in CISCO

Numbered list

```
RXY(config)# interface fa0/0
RXY(config-if)# ip access group 1 ?
      in      inbound packet
      out     outbound packet
RXY(config-if)# ip access group 1 in
RXY(config-if)# no ip access-group 1 in
RXY(config-if)# end
```

Named list

```
RXY(config-if)# ip access-group NAMED_ACL in
```

QoS quality of service

It enables the parallel use of services:

- queuing,
- packet coloring,
- bandwidth limiting (rate-limiting, shaping, policing),
- by means of mechanisms available in network protocols (eg Frame Relay, ATM, MPLS, etc.).

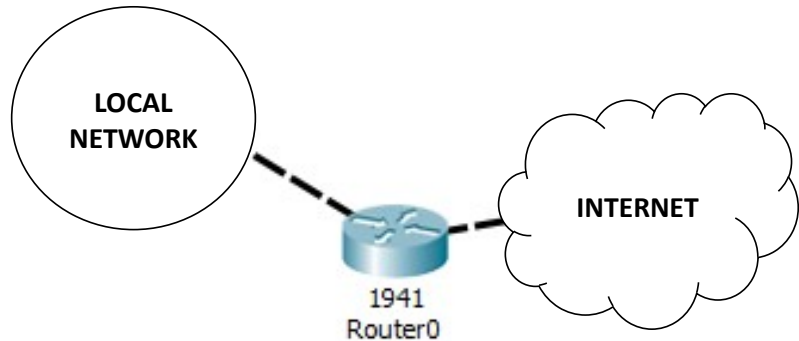
NAT

Local addresses

Class	IP address range	Net mask	CIDR notation
A	10.0.0.0 – 10.255.255.255	255.0.0.0	10.0.0.0/8
B	172.16.0.0 – 172.31.255.255	255.240.0.0	172.16.0.0/12
C	192.168.0.0 – 192.168.255.255	255.255.0.0	192.168.0.0/16

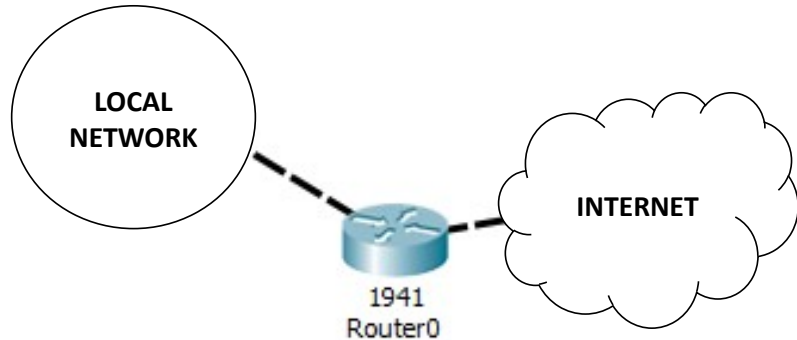
Default gateway

Default gateway is a router distinguished in the host configuration that enables the forwarding of IP packets outside the local network



NAT

NAT (Network Address Translation) – mapping private network addresses to public network addresses. Thanks to NAT, local networks can access the Internet from a public IP address.



Dynamic translation vs Pat

Types of NAT

1. Static

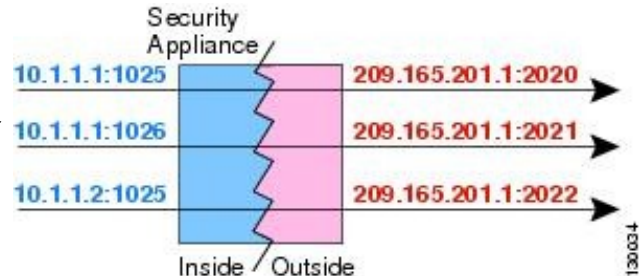
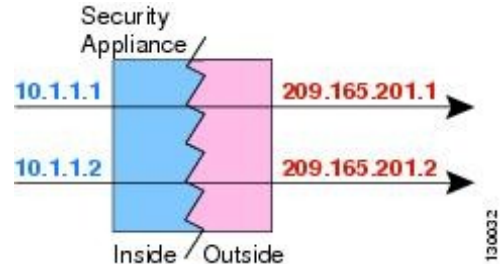
- One private IP address to one public IP address

2. Dynamic NAT

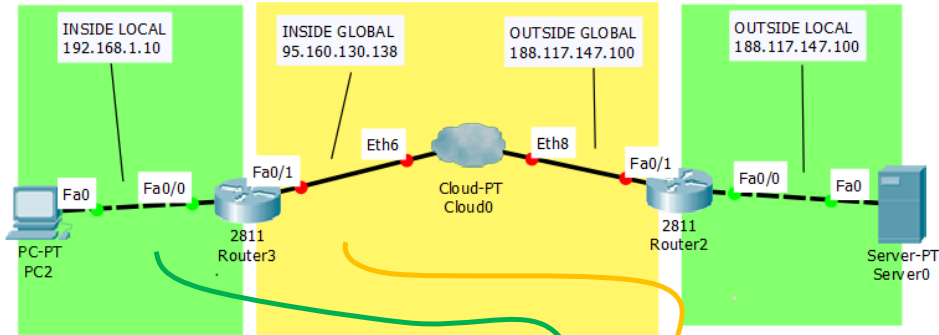
- Each private IP address to one public IP address

3. NAT with overload (PAT)

- Many private IP addresses share one public IP address



NAT – principle of operation and configuration



```
Router# configure terminal
Router(config)# interface fastEthernet 0/0
Router(config-if)# ip nat inside
Router(config-if)# interface fastEthernet 0/1
Router(config-if)# ip nat outside
Router(config-if)# exit
```

A green line connects the `ip nat inside` command to the `Router3 Fa0/0` interface in the diagram. An orange line connects the `ip nat outside` command to the `Router3 Fa0/1` interface in the diagram.

1. Static NAT – static translation

❖ One-to-one static NAT

```
Router(config)# ip nat inside source static 192.168.10.1  
200.20.10.1
```

❖ One-to-one static NAT with port

```
Router(config)# ip nat inside source static tcp 192.168.40.1 80  
200.20.10.4 80
```

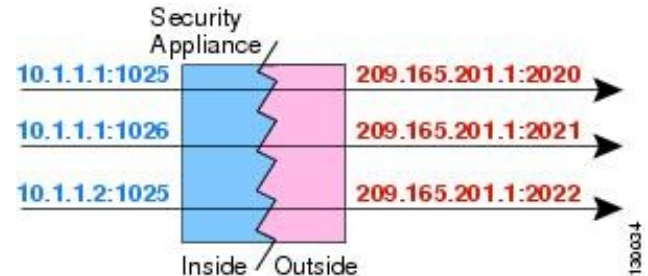
3. PAT - overloaded NAT translation

Multiple Addresses Assignment for Internal Networks to one public address

```
Router(config)# access-list 1 permit 192.168.20.0 0.0.0.255
```

```
Router(config)# ip nat inside source list 1 int fa0/1 overload
```

The combination of an IP address and port number is called a **socket**
10.1.1.2:1025



NAT – analysis of NAT operation in CISCO

Router# sh ip nat statistics

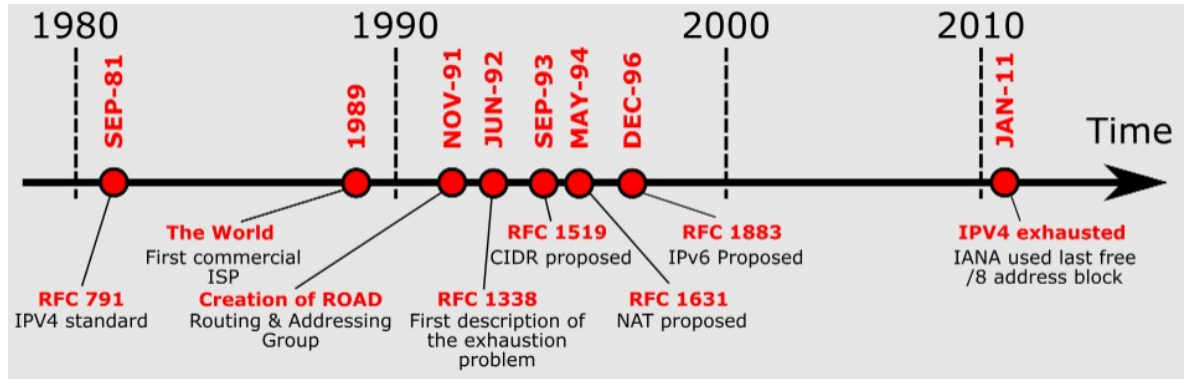
```
Router#sh ip nat st
Router#sh ip nat statistics
Total translations: 8 (1 static, 7 dynamic, 7 extended)
Outside Interfaces: GigabitEthernet0/0/1
Inside Interfaces: GigabitEthernet0/0/0.10 , GigabitEthernet0/0/0.20
Hits: 5 Misses: 7
Expired translations: 0
Dynamic mappings:
```

Router# sh ip nat translations

```
Router#sh ip nat translations
Pro  Inside global      Inside local      Outside local     Outside global
icmp 1.0.0.1:1024        172.16.20.2:2    1.0.0.2:2        1.0.0.2:1024
icmp 1.0.0.1:1025        172.16.20.2:3    1.0.0.2:3        1.0.0.2:1025
icmp 1.0.0.1:1026        172.16.20.2:4    1.0.0.2:4        1.0.0.2:1026
icmp 1.0.0.1:1         172.16.10.2:1    1.0.0.2:1        1.0.0.2:1
icmp 1.0.0.1:2         172.16.10.2:2    1.0.0.2:2        1.0.0.2:2
icmp 1.0.0.1:3         172.16.10.2:3    1.0.0.2:3        1.0.0.2:3
icmp 1.0.0.1:4         172.16.10.2:4    1.0.0.2:4        1.0.0.2:4
---  1.0.0.1           172.16.10.2      ---              ---
```

Network socket

The pool of IP addresses is only $2^{32} = 4,294,967,295$ addresses!
Thanks to PAT translation, the Internet on IPv4 still exists!



https://en.wikipedia.org/wiki/IPv4_address_exhaustion

IP Masquerade

IP Masquerade (masking IP addresses)

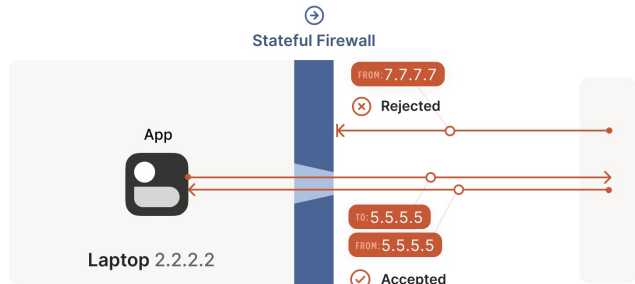
- hiding (masking) IP addresses of computers on the network by a router connected to the Internet (thanks to NAT)
- NAT-PAT allows you to convert multiple IP addresses to another single IP address 😊

NAT traversal (NAT-T)

Definition:

- a set of techniques used to establish a connection between two devices behind different NATs or firewalls.
- often used in P2P applications to enable direct communication between devices on different networks,
- implemented in VPNs, such as IPsec and VoIP techniques

<https://tailscale.com/blog/how-nat-traversal-works/>

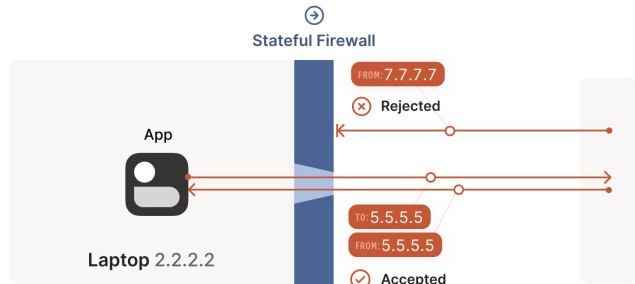


NAT traversal (NAT-T)

Configuration of NAT Traversal (NAT-T):

- Opening a NAT socket by sending a UDP message from the outside,
- Using an external coordinator server that will inform clients behind the NAT about their addressing (UDP hole punching),
- Cyclic sending of UDP messages to maintain the connection with the server (Keepalive packets)

<https://tailscale.com/blog/how-nat-traversal-works/>



NAT vs PROXY

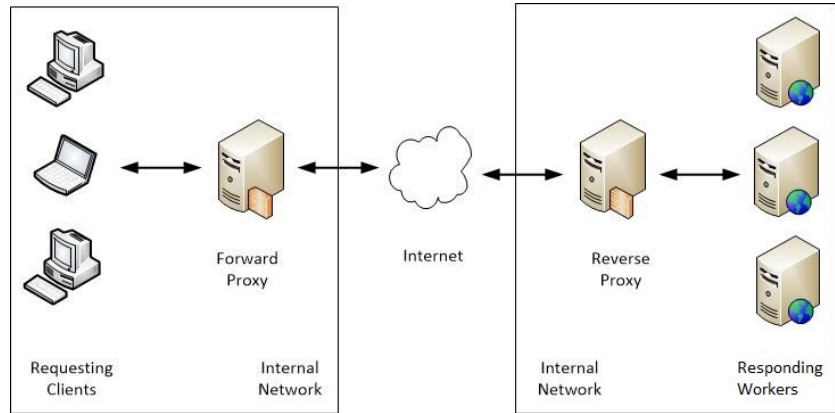
- NAT works at the network layer, while proxy works at the application layer.
- NAT is transparent to different applications. Proxy must refer to the IP address of the proxy server specified in the applications.

	Proxy	NAT
ISO layer	7	3/4
Depends on the protocol	Mainly HTTP, FTP etc.	Depends on IP addresses
Software	Needed	Not needed
Visibility	client knows about presence of a proxy (IP address of proxy server)	Transparent for the client

Forward, reverse proxy

Proxy - an intermediary between a client requesting a resource and a server providing that resource:

- Forward proxy
- Reverse proxy



Forward, reverse proxy

Forward proxy

- handles an outgoing request from a client, forwards it to the Internet, and returns a generated response to the client
- enables connections outside the network without compromising the security of the internal network

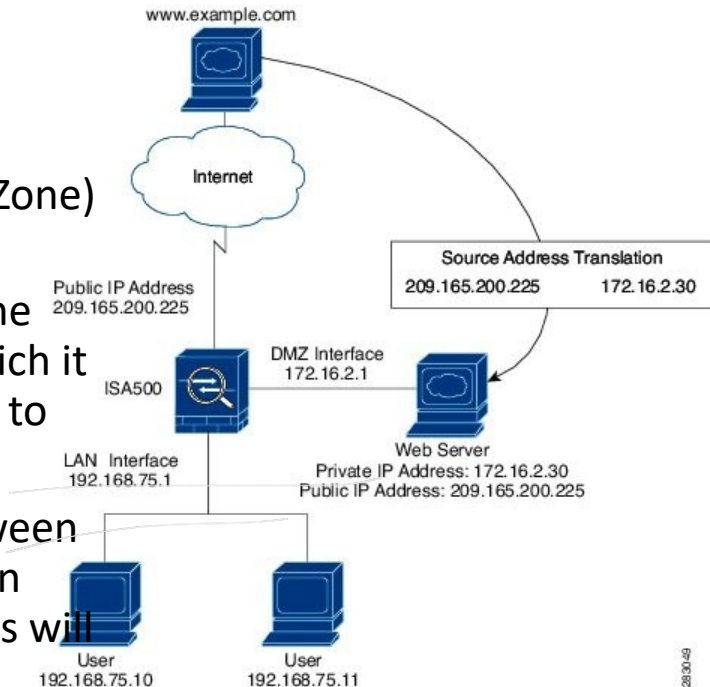
Reverse proxy

- An application that sits in front of back-end applications and forwards client requests (such as a browser) to those applications
- Client does not know the destination
- Open source web servers such as Apache, Nginx, and Caddy

DMZ

DMZ (Demarcation Zone, Demilitarized Zone)

- a segment separated from the local network, to which connections from the Internet are still possible, but from which it is impossible to establish a connection to the local network
- Implementation by routing traffic between the LAN and the DMZ through the main firewall, on which the appropriate rules will be defined



https://www.cisco.com/c/dam/assets/sol/sb/isa500_emulator/help/guide/ad1681599.html

STP, RSTP, CDP

STP - Spanning Tree Protocol

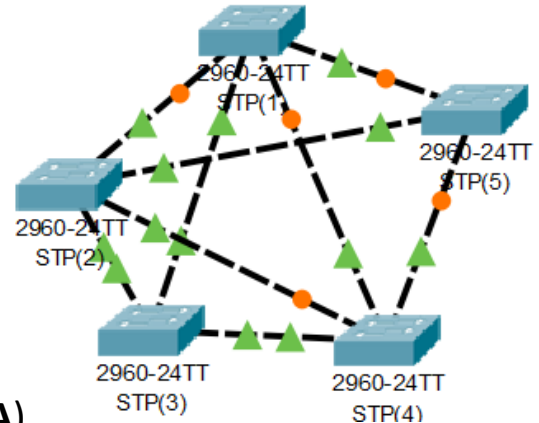
STP (Spanning Tree Protocol)

enable the configuration of switches in a way that prevents the formation of loops (broadcast storms) of layer 2 of ISO / OSI models.

Types of STP ports

- Designated,
- Main,
- Alternative,

Implementuje algorytm drzewa rozpinającego (**STA**).



STP - Spanning Tree Protocol

Switch(config)# show spanning-tree

```
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0001.4381.A671
             Cost        19
             Port        4(FastEthernet0/4)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

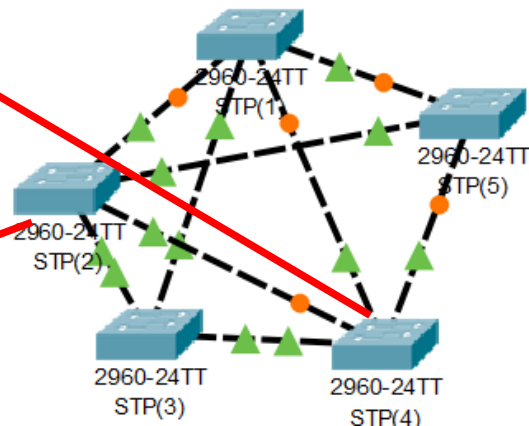
  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     0060.3EB8.926D
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time   20
```

```
Switch#sh span
VLAN0001
  Spanning tree
```

```
Root ID    P
A
T
H
Bridge ID  P
```

```
Interface  Role Sts Cost      Prio.Nbr Type
-----
Fa0/3      Altn BLK 19       128.3    P2p
Fa0/2      Altn BLK 19       128.2    P2p
Fa0/1      Altn BLK 19       128.1    P2p
Fa0/4      Root FWD 19       128.4    P2p
```

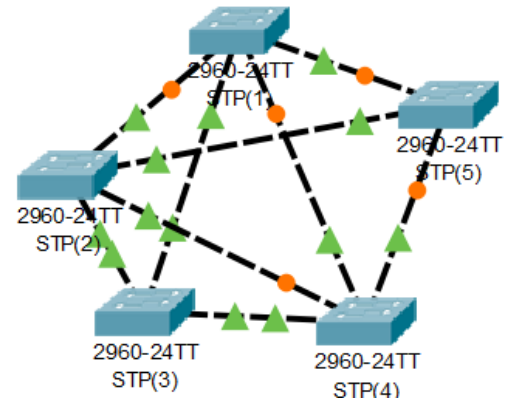
```
Fa0/2      Desg FWD 19       128.2    P2p
Fa0/1      Desg FWD 19       128.1    P2p
Fa0/3      Desg FWD 19       128.3    P2p
Fa0/4      Desg FWD 19       128.4    P2p
```



RSTP – Rapid Spanning Tree Protocol

Amendment introduced to the STP protocol (according to 802.1w standards). RSTP improves the operation of the STP protocol by providing mechanisms for rapid network convergence.

`Switch(config)# spanning-tree mode rapid-pvst`



CDP

CDP (Cisco Discovery Protocol) – is a protocol that allows you to create a base image of the nearest neighborhood in a network.

- layer II (data link) in the OSI model,
- CISCO Company Protocol,

RXY(config)# show cdp neighbors



```
Switch#  
Switch#sh cdp ne  
Capability Codes: R - Router, T - Trans Bridge, B - Sour  
S - Switch, H - Host, I - IGMP, r - Re  


| Device ID | Local Intrfce | Holdtme | Capability | Pla |
|-----------|---------------|---------|------------|-----|
| Switch    | Fas 0/1       | 136     | S          | 296 |
| Switch    | Fas 0/2       | 136     | S          | 296 |
| Switch    | Fas 0/4       | 136     | S          | 296 |
| Switch    | Fas 0/3       | 136     | S          | 296 |

  
Switch#
```

Sources

1. Akademia sieci CISCO Exploration;
Semestr 1 Podstawy Sieci, Mark A. Dyne;
2. Akademia sieci CISCO Exploration;
Semestr 2 Protokoły i koncepcje routingu, Rick Graziani;
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3 Przełączanie sieci LAN i sieci bezprzewodowe, Wayne Lewis;
4. CCNA 200-125 Zostań administratorem sieci komputerowych,
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